

Homework 2

Interconnect Topologies and Parallelism

Introduction

This assignment extends the study of collective communication in distributed deep learning to more complex GPU topologies and parallelism strategies. Building on Homework 1, **tensor parallelism** is introduced as a new dimension alongside data parallelism. Where data parallelism splits the input data across GPUs, tensor parallelism instead partitions the model parameters themselves across devices.

The two questions below ask how the choice of interconnect topology, combined with these parallelism strategies, shapes the minimum communication time when operating a single fully-connected layer. Support each answer with a step-by-step derivation for both the forward and backward passes.

Note: In the forward pass, each output activation only needs to reside on the GPU that will use it to compute the gradient of the corresponding input activation. Similarly, in the backward pass, each GPU only needs to obtain the gradients of the weights and input activations it holds.

Question 1. Crossbar Interconnect: Data & Tensor Parallelism (8 pts)

Consider a **crossbar interconnect** between 8 GPUs. Figure 1 shows a representative example. Each GPU has an inbound and outbound bandwidth of 64 GiB/s. Data and models reside entirely in GPU memory, and all values (activations, parameters, and gradients) use 32-bit floating-point precision.

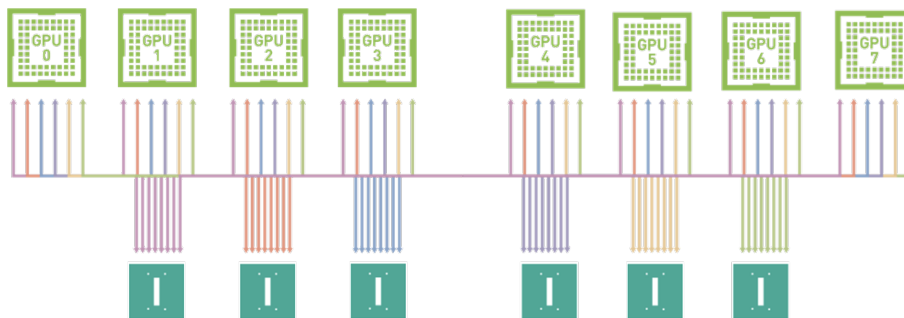


Figure 1: Crossbar interconnect topology with 8 GPUs.

The workload is a fully-connected (FC) layer with 2^{17} input activations and 2^{17} output activations, and an effective batch size of 1024.

- What is the minimum communication cost in milliseconds of the FC layer in a **data-parallel** implementation that uses **decentralized all-reduce** operations? (3 pts)
- What is the minimum communication cost in milliseconds of the FC layer in a **tensor-parallel** implementation that uses **decentralized all-gather** and **all-reduce** operations? (5 pts)
- Support your answers with full derivations for both the forward and backward passes.

Question 2. 2D Torus Interconnect: Combined Data & Model Parallelism (12 pts)

Consider a **2D Torus interconnect** with 16 GPUs arranged in 4 rows and 4 columns as illustrated in Figure 2. All links are **unidirectional** and have a bandwidth of 16 GiB/s. As in the previous question, data and models reside in GPU memory, and all values (activations, parameters, and gradients) use 32-bit floating-point precision.

The workload is again a fully-connected layer with 2^{17} input activations and 2^{17} output activations, and an effective batch size of 1024.

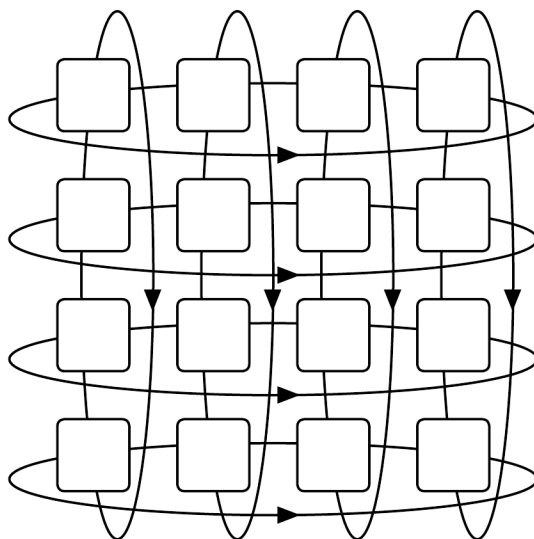


Figure 2: 2D Torus topology with 16 GPUs (4 rows $\times$ 4 columns).

The goal is to exploit **data parallelism** and **model parallelism** simultaneously:

- **Data parallelism:** each row of the 2D Torus processes 256 of the 1024 input samples.
- **Tensor parallelism:** each column computes 2^{15} output activations per input sample.

What is the minimum communication cost in milliseconds of the FC layer that uses **ring all-reduce** and **ring all-gather** operations in this setting? Support your answer with the full derivations for both the forward (**4 pts**) and the backward (**8 pts**) passes.